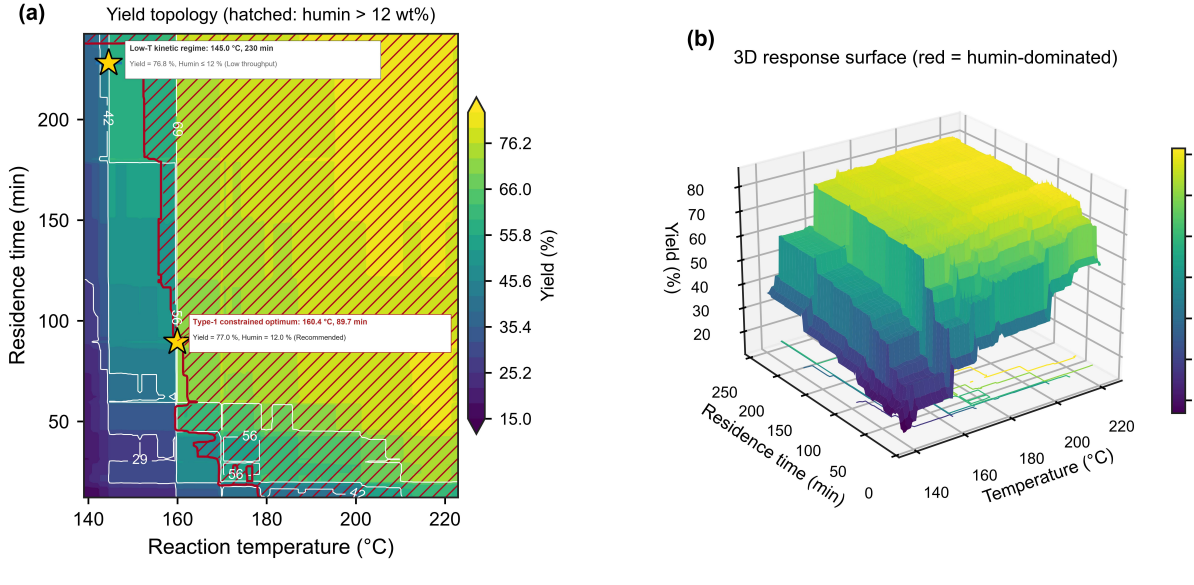

(a)
Yield topology (hatched: humin > 12 wt%)
Low-T kinetic regime: 145.0 °C, 230 min
Yield = 76.8 %, Humin ≤ 12 % (Low throughput)
Type-1 constrained optimum: 160.4 °C, 89.7 min
Yield = 77.0 %, Humin = 12.0 % (Recommended)
42
89
36
4
56
29
56
42
76.2
66.0
55.8
45.6
35.4
25.2
15.0
Yield (%)
Residence time (min)
Reaction temperature (°C)
140
160
180
200
220
50
100
150
200
(b)
3D response surface (red = humin-dominated)
Yield (%)
80
70
60
50
40
30
20
Residence time (min)
250
200
150
100
50
0
Temperature (°C)
140
160
180
200
220